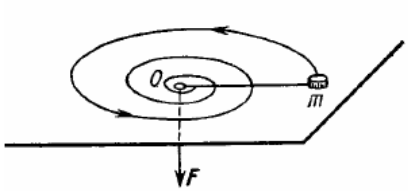
PHY101: Introduction to Physics-1

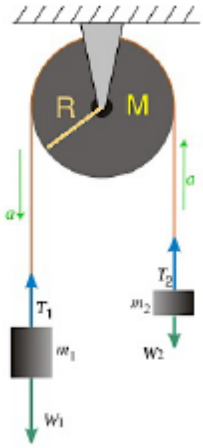
Monsoon, 2018

Tutorial-11

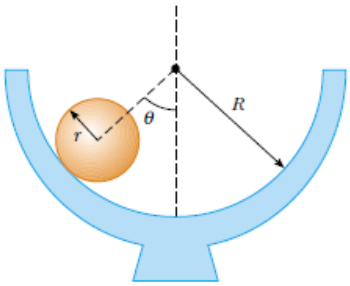
1. A small body of mass m tied to a non-stretchable thread moves over a smooth horizontal plane. The other end of the thread is being drawn into a hole 0 (see figure) with a constant velocity. Find the thread tension as a function of the distance r between the body and the hole if at $r = r_0$ the angular velocity of the thread is equal to ω_o.



2. Consider an Atwood machine with blocks of mass m_1, m_2 (with $m_1 > m_2$), respectively. The pulley is of mass M and radius R; see the adjoining figure. Find out the acceleration of the blocks and tensions in the string. The rope is assumed to be massless and no slipping of the rope occurs over the pulley. Express your results in terms of m_1, m_2, M, R and g where g is the acceleration due to gravity.



3. A uniform solid sphere of radius r is placed on the inside surface of a hemispherical bowl with much larger radius R. The sphere is released from rest at an angle θ to the vertical and rolls without slipping (see figure). Determine the angular speed of the sphere when it reaches the bottom of the bowl.



4. Without the wheel, a bicycle frame has a mass of 8.44 kg. Each of the wheels can be roughly modeled as a uniform solid disk with mass of 0.82 kg and a radius of 0.343 m. Find the kinetic energy of the whole bicycle when it is moving forward at 3.35 m/s.